Tiago Autonomous Exploration and Manipulation

Autonomous and Mobile Robotics - Project

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Overview

1. Introduction
2. Task 1: Exploration
3. Task 2: Navigation
4. Task 3: Manipulation

Project Overview

- **Objective:** Develop a complete autonomous system for the TIAGo robot combining **navigation** and **manipulation** skills
- **Tasks:**
  1. Map Generation (SLAM)
  2. Autonomous Localization & Exploration
  3. Pick and Place of specific ArUco cubes

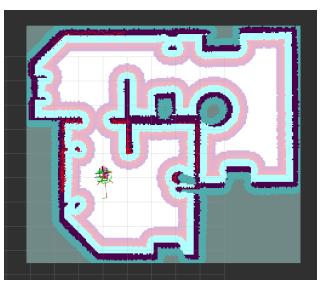


Figure 1: Map Generation in RViz2

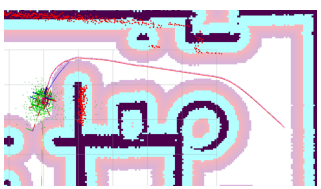


Figure 2: Autonomous Localization & Exploration

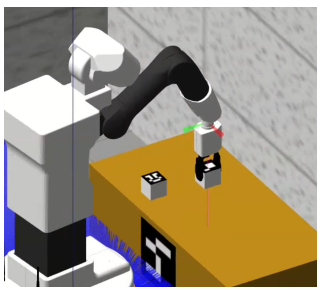


Figure 3: Pick and Place of specific ArUco cubes

System Architecture & Design

Development Stack

- **Framework:** ROS2 Humble Hawksbill
- **Simulation:** Gazebo 11 (Physics & Environment)
- **Visualization:** RViz2 (Real-time sensor data & markers)

Architectural Pattern

We adopted a modular **Policy vs. Actuator** approach:

- **Main Policy (Coordinators):** High-level nodes managing the task logic and state machine
- **Generalized Actuators:** Reusable skills (e.g., `map_navigator`, `head_motion`) agnostic of the specific task
- **Handshake Protocol:** Coordination is handled via strictly defined Topics, Services, and Actions to ensure robust synchronization[a]

[a] Custom messages and service in a **ament_cmake** package: `tiago_exam_interfaces`.

Package Organization: tiago_exam

The Project is entirely accomplished inside the tiago_exam package. This is a **ament_python** package structured as follows:

```
tiago_exam/
├── launch/ ..... Launch files for task environments
├── tiago_exam/ ..... Sub-packages and shared utilities
├── maps/
│   ├── group29_map.yaml
│   └── group29_map.pgm
├── data/ ..... CSV records for ArUco markers
├── config/ ..... RViz configs and map data
└── temp/ ..... Debug images for exploration
```

Task 1: Autonomous Map Generation

Objective

Perform a **full autonomous exploration** of **an unknown environment** to generate a consistent 2D occupancy grid map using SLAM

Environmental Challenges

The testing environment presents **specific difficulties** that standard configuration[a] could not handle:

- **Complex Geometry:** Narrow passages and obstacles requiring precise maneuvering
- **Lighting Artifacts:** Strong shadows created false positive obstacles in the LIDAR readings (ghost objects)

[a] Default SLAM and navigation parameters

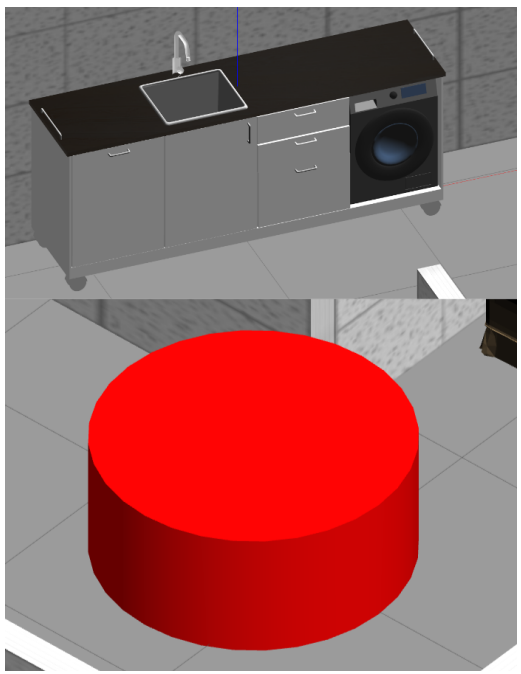


Figure 4: Environment in Gazebo 11

Task 1: Exploration Strategy

To achieve efficient coverage, we integrated a C++ node based on `m-explore`, enhancing the standard **frontier exploration logic**.

Optimized Frontier Selection Logic

We tuned the cost function parameters to prioritize frontiers that are:

1. **Closer** to the robot's current pose (minimizing travel time)
2. Located in **larger unexplored voids** (maximizing information gain)

Completion Workflow

Upon exhaustion of the whitelist frontiers, the robot automatically navigates back to the **Home Position** and triggers the **map saver node**, storing the result in the `maps/` directory.

Task 1: Safety Configuration

Safety Measure: Pre-Exploration Configuration

Before moving, the system executes a mandatory **Tuck Arm** procedure. Folding the arm into a safe configuration minimizes the robot's footprint, preventing collisions with obstacles during autonomous navigation.

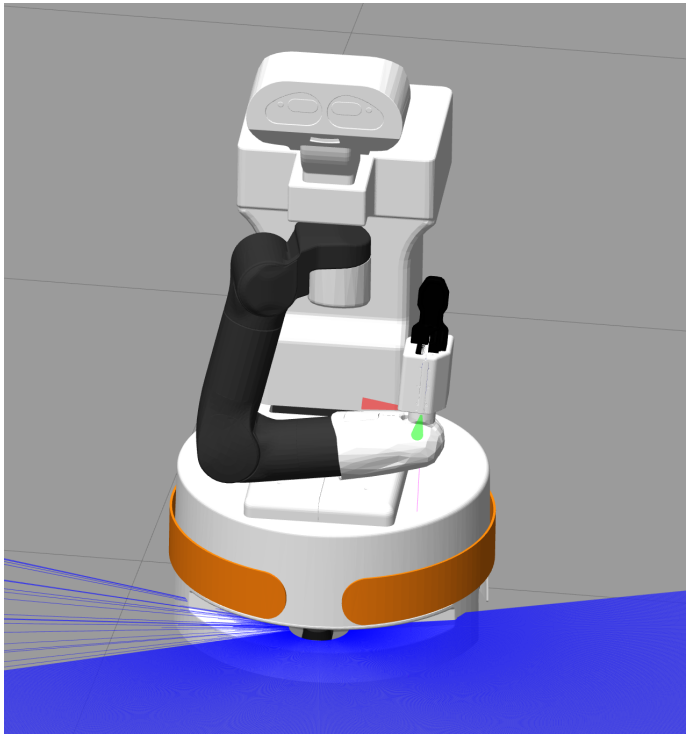


Figure 5: Tuck Arm Configuration for Safe Navigation

This configuration is also used for Task 2 and Task 3 navigation.

Task 1: Navigation Stack Architecture

Critical System Enhancements

- **Controller Logic Injection:** Implemented `RotationShimController` to enforce precise in-place rotation, eliminating erratic spiraling in tight spaces
- **Map Integrity Verification:** Tightened SLAM loop-closure thresholds to reject weak scan matches and prevent long-term map corruption
- **Safety Gradient Tuning:** Adjusted the inflation decay to create a steeper potential field, actively repelling the planner from obstacles
- **Artifact Clearing:** Extended raytracing and enforced height thresholds to aggressively filter floor reflections and "ghost" obstacles

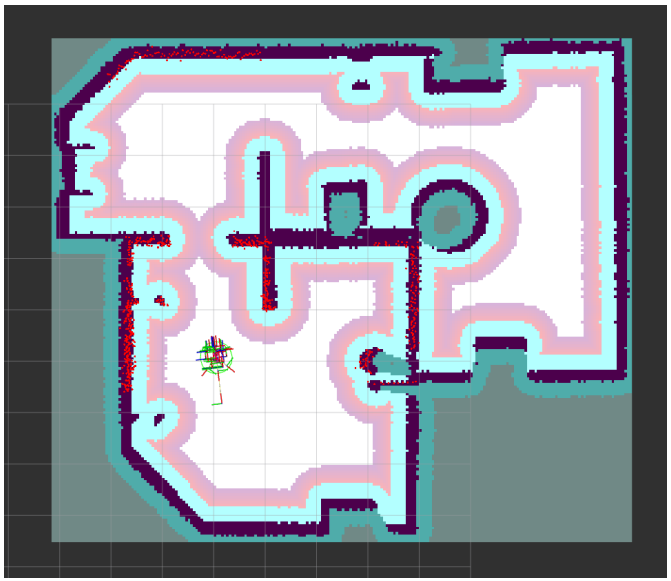


Figure 6: Optimized Occupancy Gradient

Task 1: Exploration Video Result

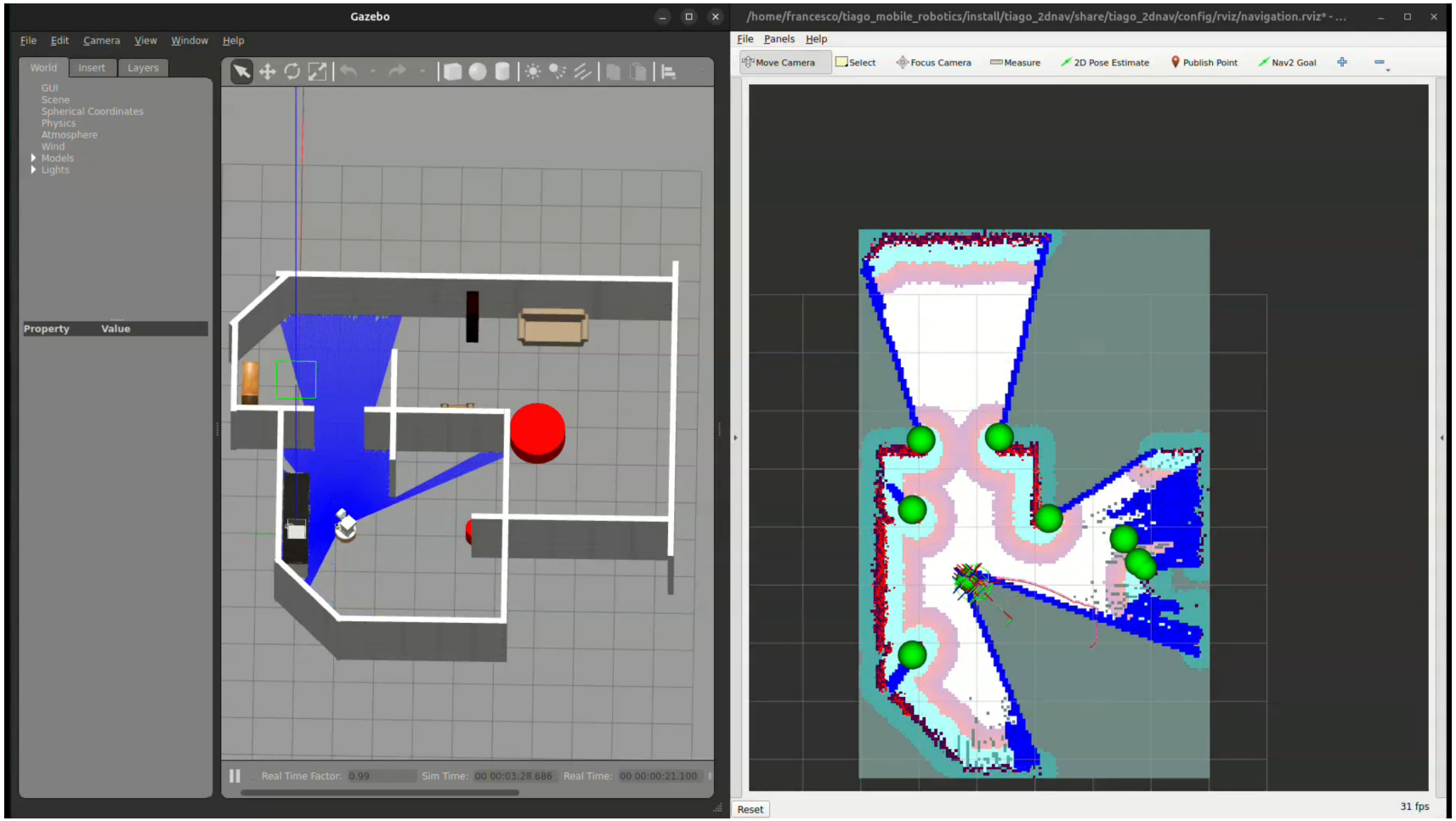


Figure 7: Exploration Task Execution (Click to play in browser)

Task 2: Autonomous Map Navigation

Objective

Perform a **full autonomous navigation** of a **known environment** using the previously generated 2D occupancy grid map to **find** and **record** the pose of specific **ArUco markers**.

- **Pick Station Marker:** Aruco Marker ID 26, size 0.25m
- **Place Station Marker:** Aruco Marker ID 238, size 0.25m

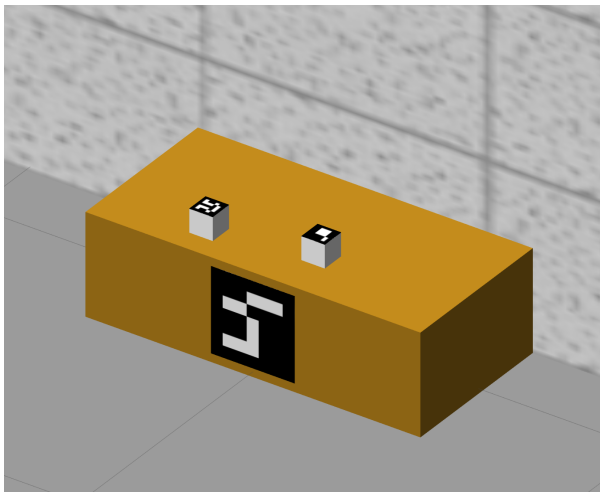


Figure 8: Pick Station Marker 26

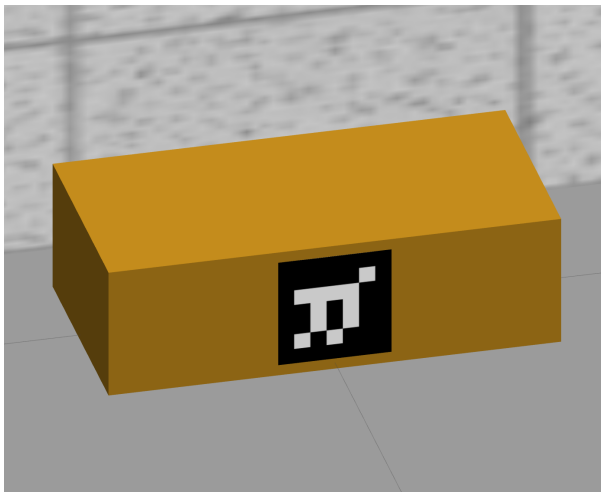


Figure 9: Place Station Marker 238

Task 2: Navigation Trade-offs

Operational Challenges

ArUco markers are attached to tables that may be **occluded** or located in **corners**.

Exhaustive Perimeter Search

To locate the tables, we evaluated an **exhaustive perimeter search** (wall-following) to systematically scan every corner.

Sampling Heuristic

However, we opted for a **uniform sampling heuristic**. By generating safe random goals within the free space, the robot covers the environment more rapidly than tracing the full perimeter, maintaining high detection reliability with lower execution time.

Task 2: Safety Navigation

Map Processing Pipeline

Using OpenCV on the occupancy grid map:

1. **Safety Mapping:** Applied `cv2.distanceTransform` to calculate the distance of every pixel to the nearest obstacle
2. **Filtering:** Created a "Whitelist Mask" of pixels where:

$$d_{obstacle} > (R_{robot} + \text{Clearance})$$

3. **Goal Selection:** Randomly samples coordinates from the whitelist and converts them from Image Frame (pixels) to World Frame (meters)

Result: The robot never attempts to navigate into walls or narrow traps.

The clearance value is set to 0.4 meters, which is a good trade-off between safety and navigable space, given the robot's dimensions and the map's obstacle density.

Task 2: System Architecture & Orchestration

The complex behavior required for Task 2 is managed by a central **Coordinator Node** that acts as the high-level decision maker.

The Coordinator Design

- **Finite State Machine (FSM):** Transition logic between *Exploring*, *Tracking*, and *Inspecting* states
- **Concurrency Handling:** Utilizes `ReentrantCallbackGroup` and `MultiThreadedExecutor`. This allows the node to process sensory interrupts (e.g., seeing a marker) immediately, even while blocking navigation actions are active
- **Interrupt Logic:** The standard exploration loop is designed to be preempted instantly upon ArUco detection

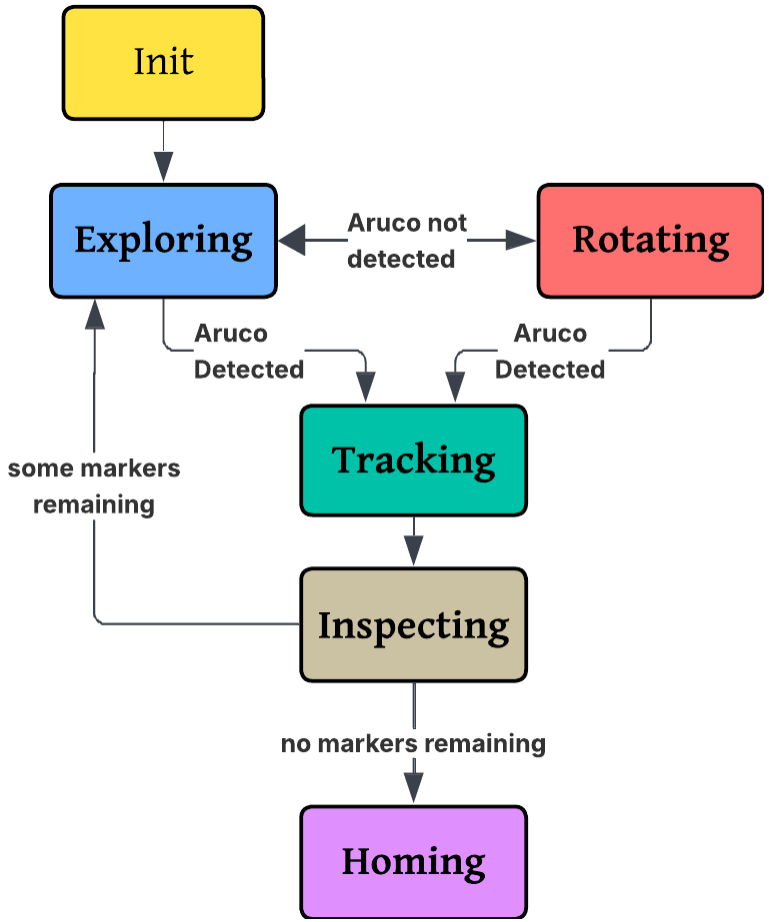


Figure 10: State Machine Logic

Task 2: Global Localization

Since the robot starts in an unknown location on the map, we implemented a **global localization** routine exploiting the Adaptive Monte Carlo Localization (AMCL).

The Converge Strategy:

1. **Particle Spread:** Trigger `/reinitialize_global_localization` to distribute AMCL particles uniformly across the map
2. **Data Gathering:** The robot executes a 360° in-place rotation at 1.0 rad/s to feed the laser scan matcher
3. **Covariance Check:** The routine continuously monitors the pose covariance matrix, ensuring it falls below strict thresholds before allowing any movement

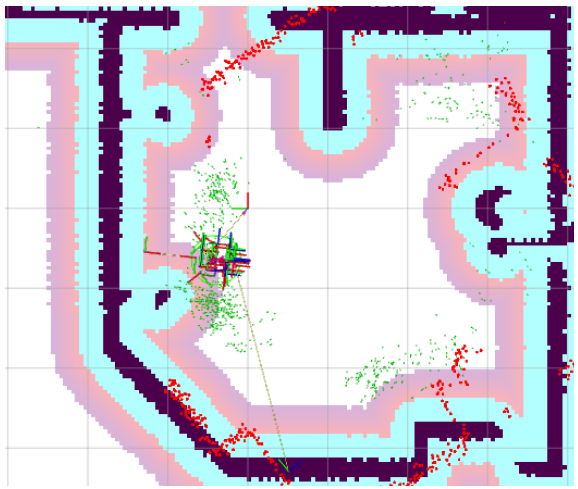


Figure 11: AMCL Particles Spread

Task 2: Global Localization Contd.

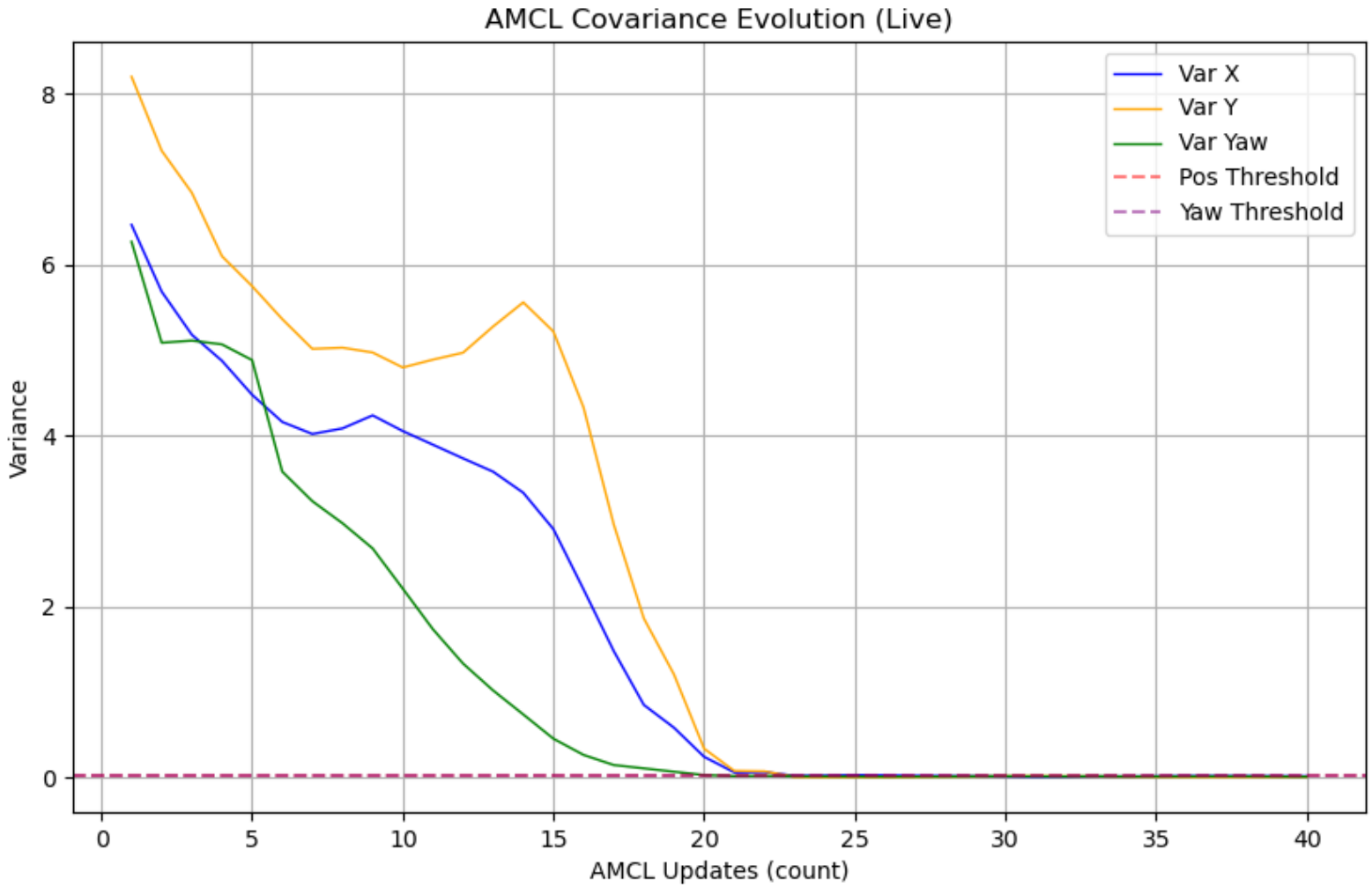


Figure 12: Covariance Reduction Over Time

Task 2: Exploration Strategy

After the localization, the Coordinator Node activates the following nodes:

Generalized Nodes

- **Aruco Marker Detector:** The node continuously monitors the camera feed for the station markers
- **Explorator:** The node processes the occupancy grid map to create a "whitelist" of safe navigation targets, ensuring the robot only attempts to navigate to locations that are free of obstacles
- **Navigator:** The robot continuously samples from the "whitelist" of safe exploration goals. After reaching each goal, the robot performs a 360° rotation to improve its environmental awareness

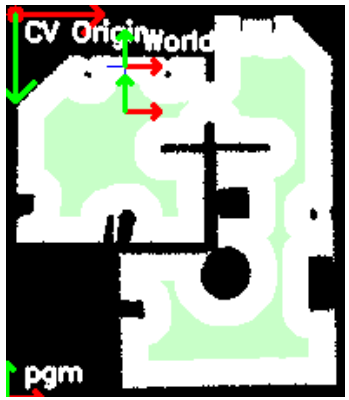


Figure 13: Clearance-Based Goal Sampling

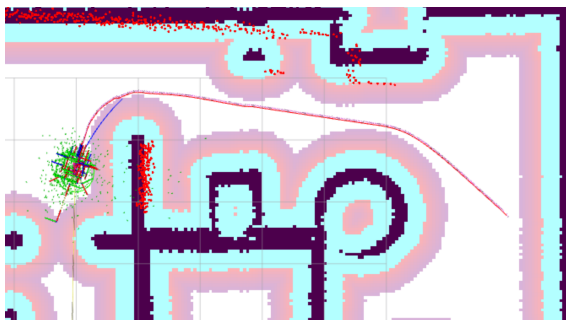


Figure 14: Random Safe Goal Navigation

Task 2: Exploration Strategy - Contd.

Exploration Strategy

- **Tracking:** Once an ArUco marker is detected, the robot is brought to a specific distance from the marker, while continuously facing it by adjusting the camera tilt and pan angles
- **Inspecting:** Once the robot is correctly positioned, it transitions to the Inspecting state, where the marker and robot Pose is saved in a `.csv` file for later use
- **Homing:** After all the markers are detected, the robot automatically navigates back to the Home Position

Task 2: Tracking Behavior

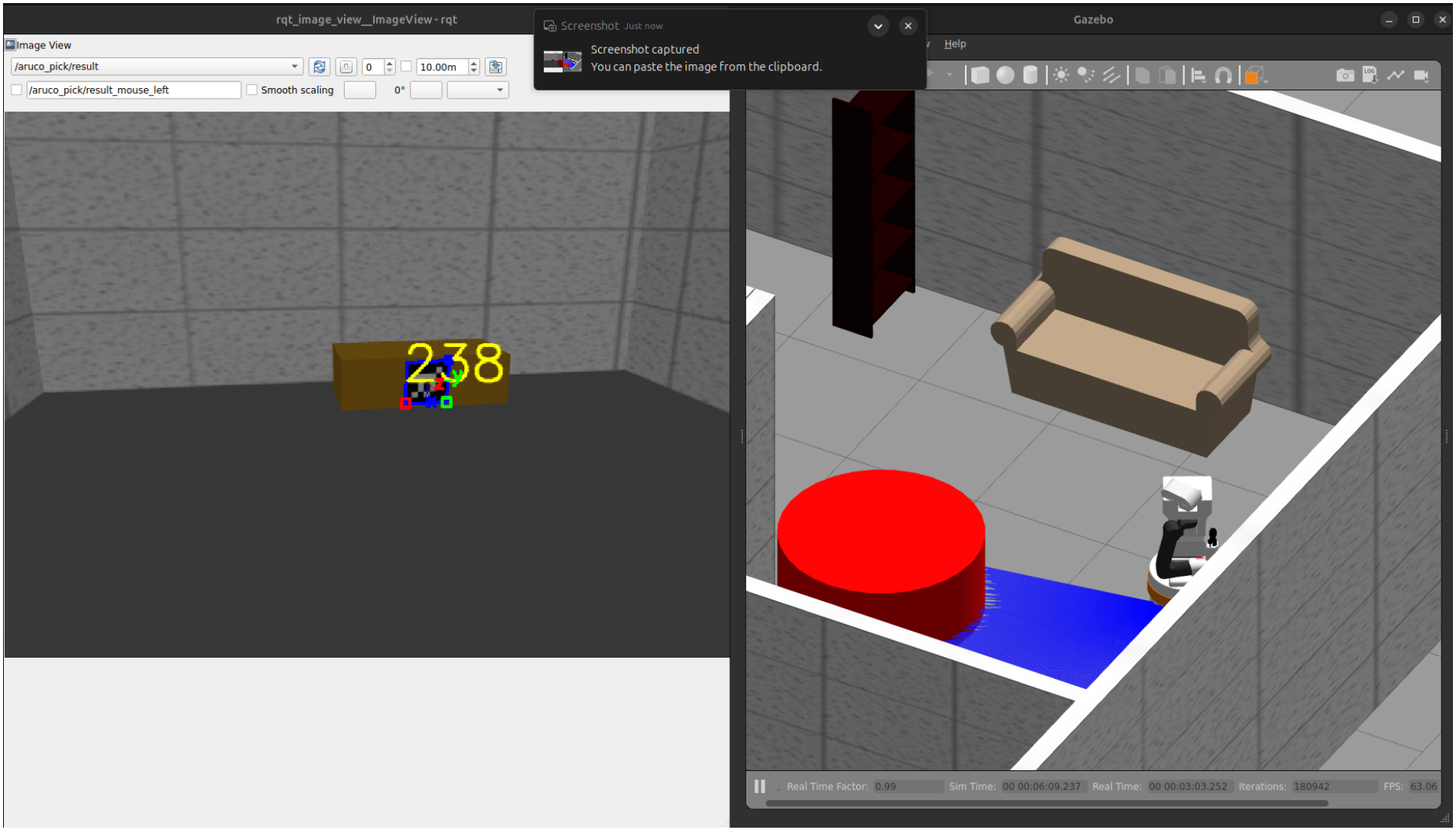


Figure 15: Aruco Tracking Behavior

Task 2: Inspecting Behavior

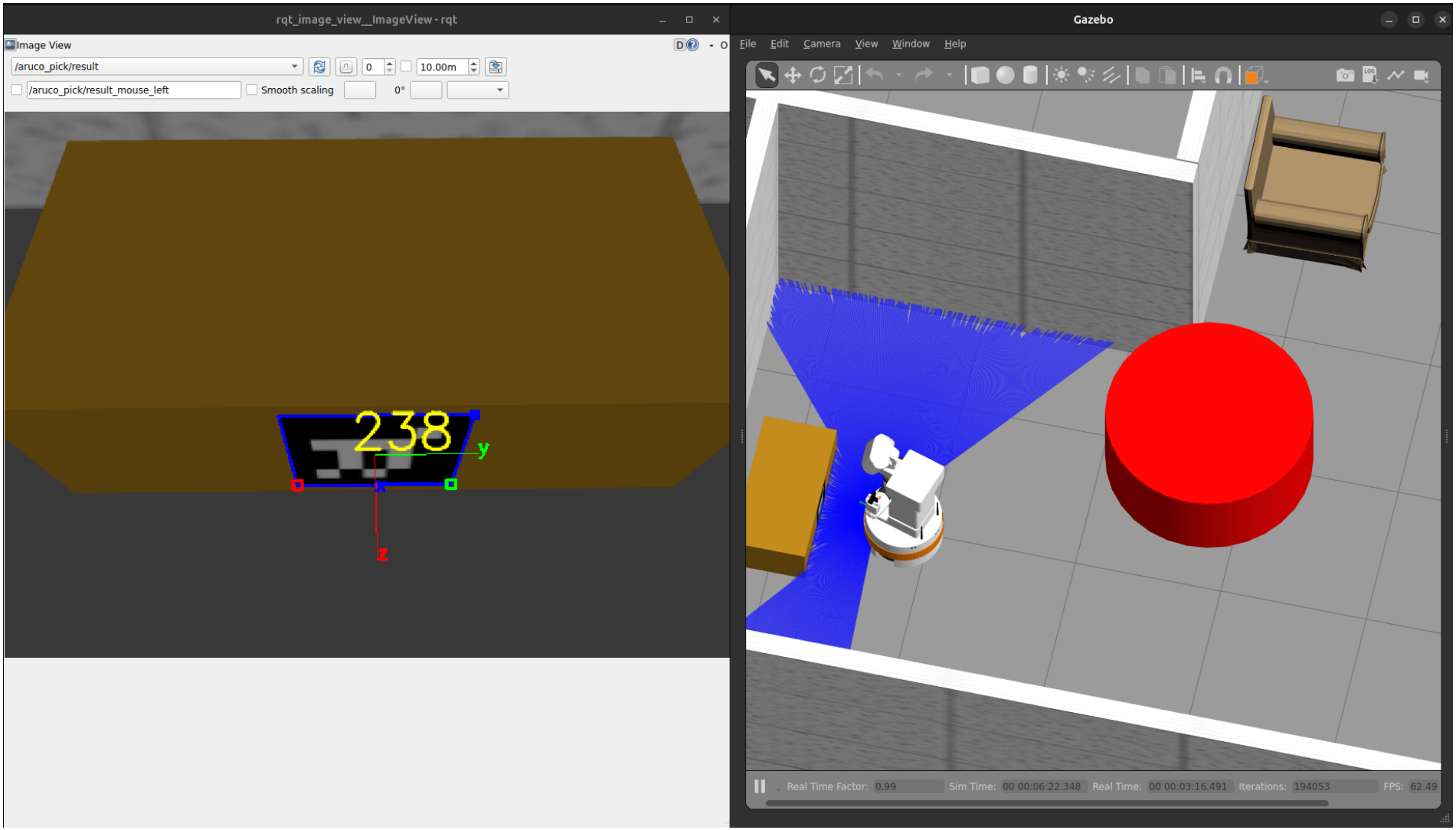


Figure 16: Aruco Inspecting Behavior

Task 2: Navigation Video Result

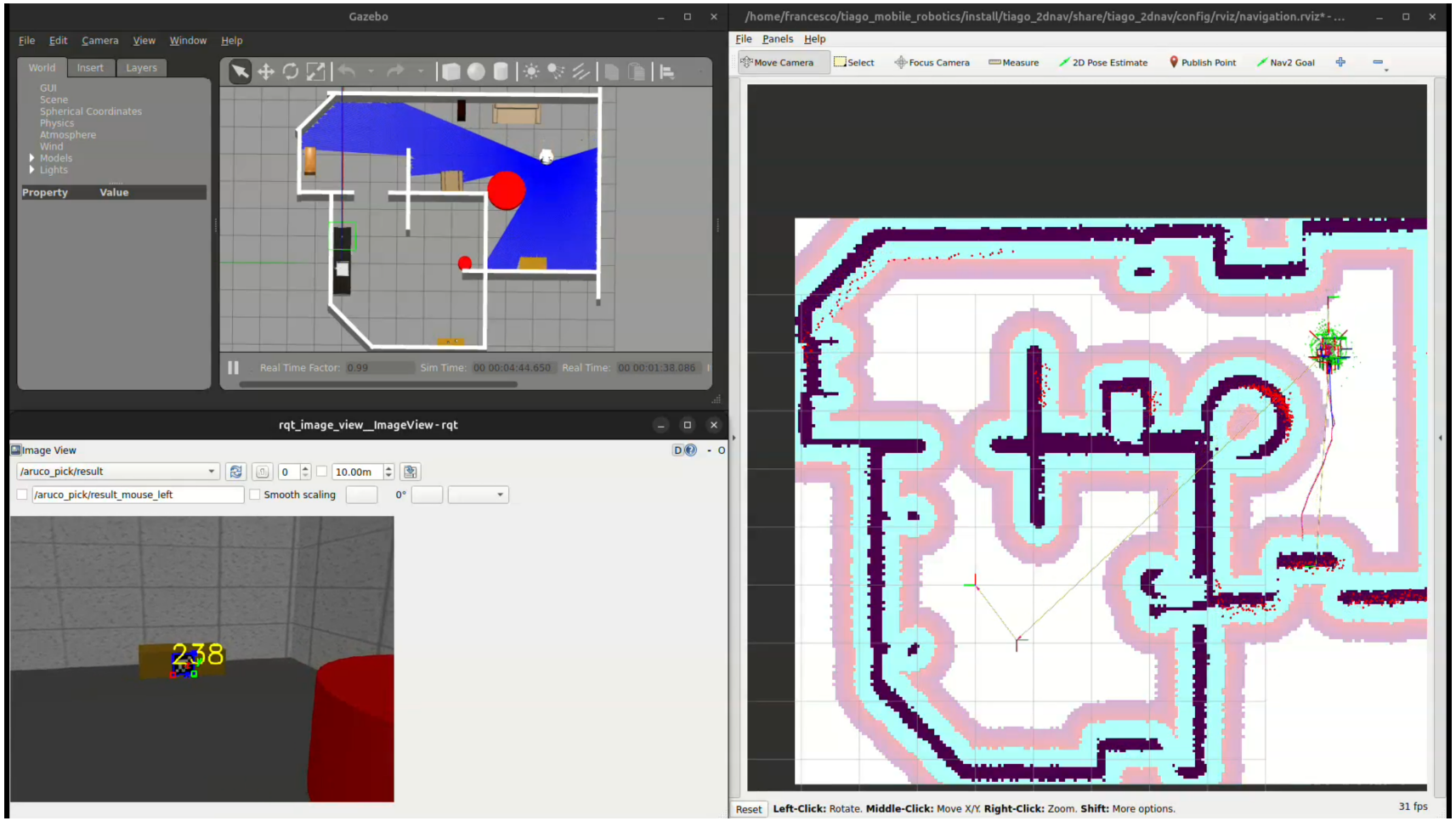


Figure 17: Navigation Task Execution (Click to play in browser)

Task 3: Pick & Place

Objective

Execute a precise **Pick & Place** operation leveraging on:

- The **map** generated in Task 1
- The **marker locations** identified in Task 2

Operational Constraints

The manipulation task requires strict **mechanical** and **spatial tolerances**:

- **Grasp Tolerance:** The parallel gripper has **minimal mechanical compliance**. Even slight ArUco pose estimation errors (*homography noise*) can cause the fingers to collide with the object rather than grasping it
- **Kinematic Reachability:** Placement success is **strictly coupled** with **Base Positioning**. If the robot stops too far from the table, the arm's maximum reach is insufficient
- **Deployment Safety:** Transitioning the arm from the *Safe Travel Configuration* to the *Work Configuration* near obstacles poses a **high risk** of **collision** with the table edge

Task 3: Coordinator

The `coordinator_task3` node orchestrates the sequence, bridging the data gathered in Task 1 and Task 2 with the actuation logic.

Execution Workflow

1. **Data Ingestion:** Parses `marker_locations.csv` to retrieve the exact pose of the tables targets found during Task 2 exploration
2. **State Sequence:** Enforces the required order of operations:

$$\text{Localize} \xrightarrow{\text{Navigate to Pick}} \text{Pick Cube ID 63} \xrightarrow{\text{Navigate to Place}} \text{Place Cube ID 63}$$

$$\text{Next Target} \xrightarrow{\text{Navigate to Pick}} \text{Pick Cube ID 582} \xrightarrow{\text{Navigate to Place}} \text{Place Cube ID 582}$$

3. **Visual Feedback:** Publishes dynamic TF frames (`target_cube_frame`) to visualize the manipulation target in RViz in real-time

Once all targets cubes are processed, the coordinator commands the robot to return to the **Home Position**, signaling Task 3 completion.

Task 3: Manipulator

The `manipulator` node orchestrates the sequence, handling the actual arm movements and gripper actions on the basis of the `coordinator_task3` commands. The commands are in the form of:

- **Pick Command:** `'PICK_<cube_id>'`
- **Place Command:** `'PLACE_<cube_id>'`

This commands issue the manipulator node to execute the corresponding manipulation sequence:

PICK_<63>

$$\text{Safe Pose} \xrightarrow{\text{Idle for aruco pose update}} \text{Pre_Grasp} \rightarrow \text{Grasp} \rightarrow \text{Post_Grasp} \rightarrow \text{Safe Pose}$$

PLACE_<63>

$$\text{Safe Pose} \xrightarrow{\text{Idle for place station}} \text{Pre_Release} \rightarrow \text{Release} \rightarrow \text{Post_Release} \rightarrow \text{Safe Pose}$$

Task 3: Coordinator & Manipulator State Machines

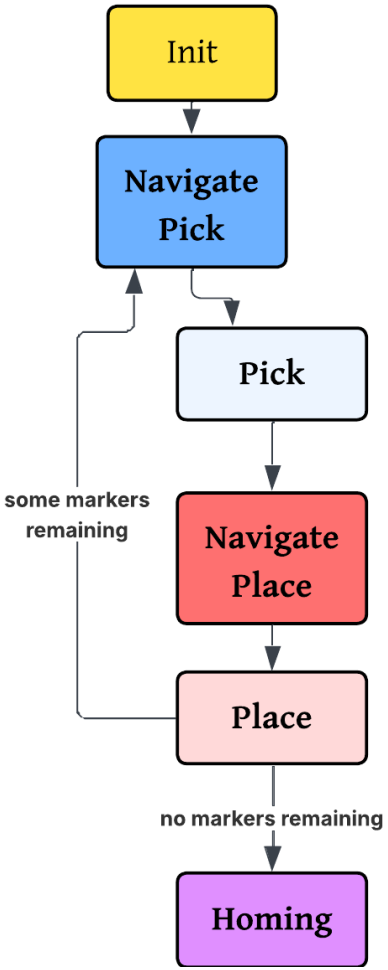


Figure 18: Coordinator

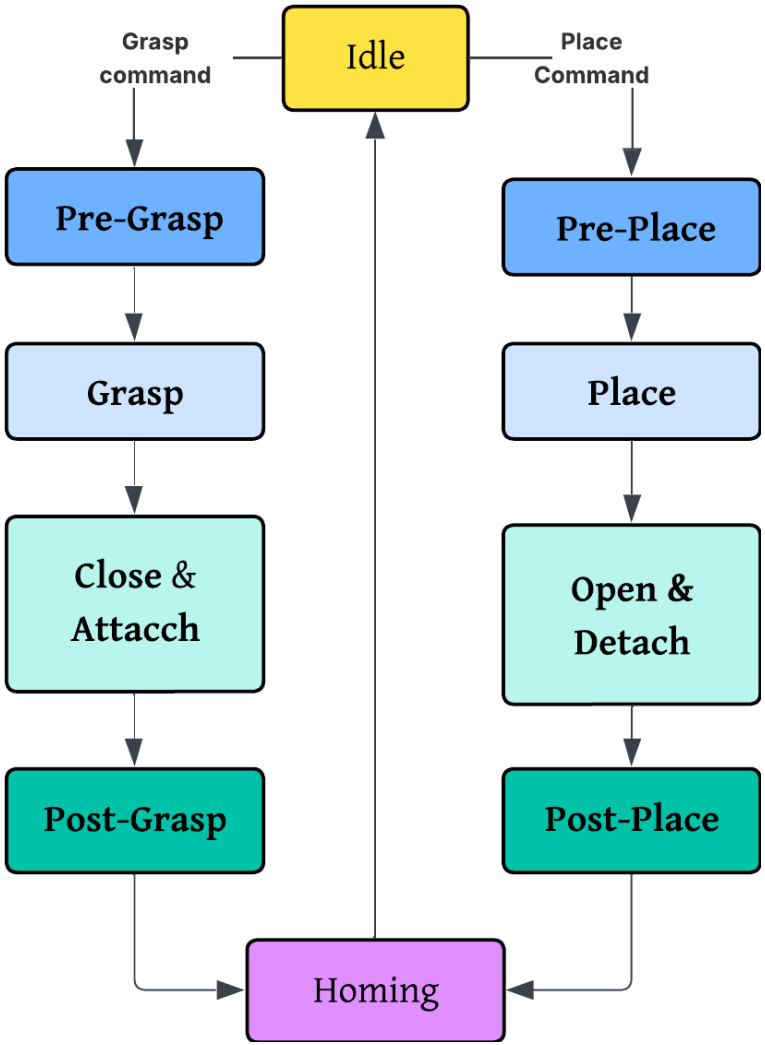


Figure 19: Manipulator

Task 3: Marker Perception

Perception Sequence

Head Scanning & Latching

- While the aruco detector is running the robot head **tilts down** when the station is reached
- The system caches the **latest valid transform**

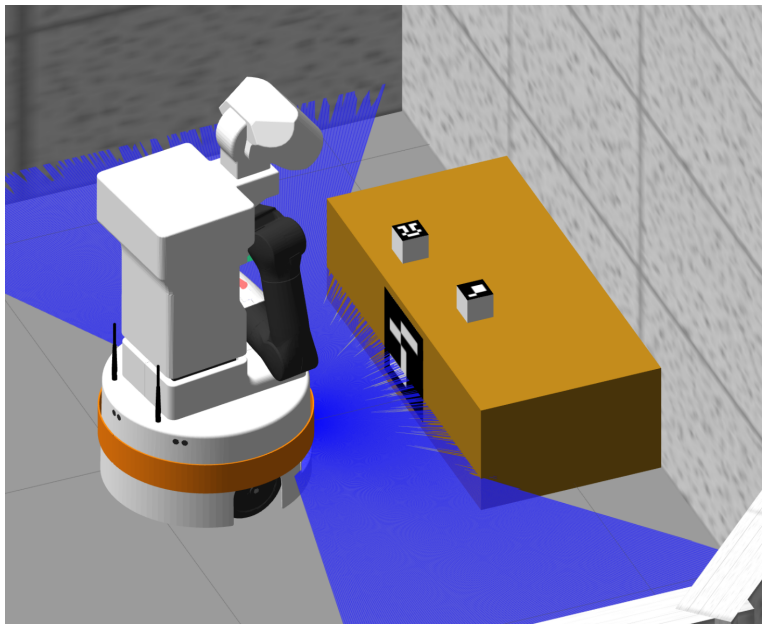


Figure 20: Head Scanning for Marker Detection

Once the robot head has latched onto the marker, the robot stares at the marker for 1 second to ensure a stable pose estimation before executing the grasping sequence.

Task 3: Grasp & Release Selection Pose

Pose Calculation

The Grasp Pose is computed by extracting the marker's transform. The choice of the grasping point has been made to ensure a stable grip on the cube, considering collinear and antipodal grasp for the 7cm cube.

Pose Computation (KDL)

Given the marker pose ${}^{base}T_{marker}$, we compute the approach vector by applying a geometrical offset:

$$ {}^{base}T_{appr} = {}^{base}T_{marker} \cdot \underbrace{T_{rotY(90^\circ)}}_{\text{Rotation}} \cdot \underbrace{T_{appr}}_{\text{Translation}} $$

This aligns the gripper normal with the cube's face.

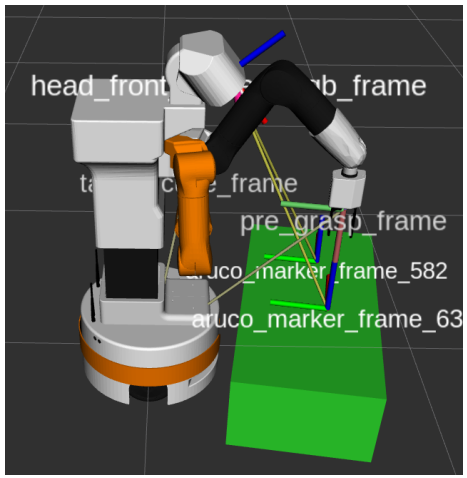


Figure 21: TF Frames visualization

Placement Logic: The release pose is derived from the grasp pose but lifted vertically to ensure the object rests on the table:

$$ P_{release_z} = P_{grasp_z} + 0.15m $$

Task 3: Manipulation Safety & Robustness

The manipulation sequence follows a strict cycle:

$$\text{Pre_op} \rightarrow \text{op} \rightarrow \text{Post_op} \rightarrow \text{Safe Pose}$$

Motion Safety: Decoupled Retraction

The $0.4m$ approach offset requires the `arm_torso` group to extend significantly. To prevent the torso from lowering while the arm is still over the table (risk of collision), we implemented a **Decoupled Safe Procedure**:

1. **Arm First:** The arm retracts to a folded configuration while the torso remains fixed
2. **Torso Second:** Only once the arm is safe, the torso lowers to the travel height

Environmental Awareness: Upon arrival, **Collision Objects** representing the table (known dimensions) are spawned in MoveIt to forbid planning through the surface.

Task 3: Manipulation Safety & Robustness (Cont.)

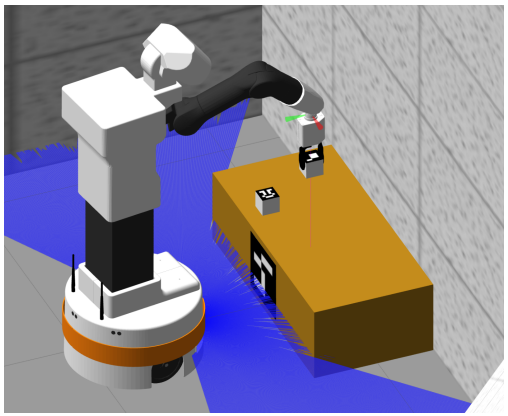


Figure 22: Cube Extraction

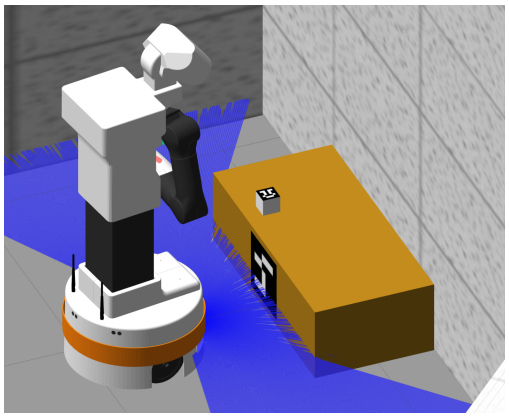


Figure 23: Arm Retraction

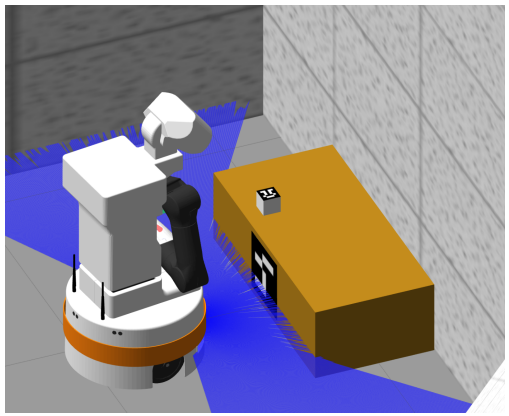


Figure 24: Torso Retraction

At the end of the manipulation operation the robot performs a *backup* maneuver to reach a safe distance from the table before rotating.

Task 3: Precision Placement Strategy

To ensure accurate object placement, we adopted a **Relative Pose Replication** strategy rather than absolute coordinate planning.

The Logic:

- The robot records the exact **base-to-table distance** (d_{ref}) during the successful Grasp phase
- This distance is enforced as a constraint during the Placement phase to guarantee the **arm's kinematic reachability**

Why?

Reusing the grasp pose relative to the base minimizes Inverse Kinematics (IK) errors and ensures the object lands exactly on the table surface.

Active Error Correction (Closed-Loop)

Since AMCL localization has a margin of error ($> 2cm$), the robot performs a final **Linear Approach/Backup** maneuver before placing:

$$e = d_{real} - d_{ref}$$

If $|e| > 1.0$cm: The base adjusts its position ($v_x = \pm 0.1$m/s) until convergence.

Task 3: Grasping Dynamics

Mobile manipulation in physics engines (Gazebo/ODE) suffers from unreliable **contact dynamics** and friction simulation, often leading to object slippage during base motion.

Attempt 1: Force Control (Failed)

We initially attempted a realistic grasp based on Force/Torque sensor feedback:

- **Tuning:** Adjusted friction coefficients (μ) and contact compliance (k_p, k_d)
- **Logic:** Stop gripper closure upon reaching a force threshold ($F_{th} = 10\text{N}$)

Outcome: Proven unreliable due to physics engine instability

Solution: Kinematic Locking

To guarantee transport stability, we bypassed the physics engine using the **Link Attacher Plugin**.

- **Mechanism:** Creates a rigid kinematic joint (Fixed Joint) between the gripper and the object
- **Result:** Zero slippage and robust transport regardless of base accelerations

Task 3: Grasping Video Result

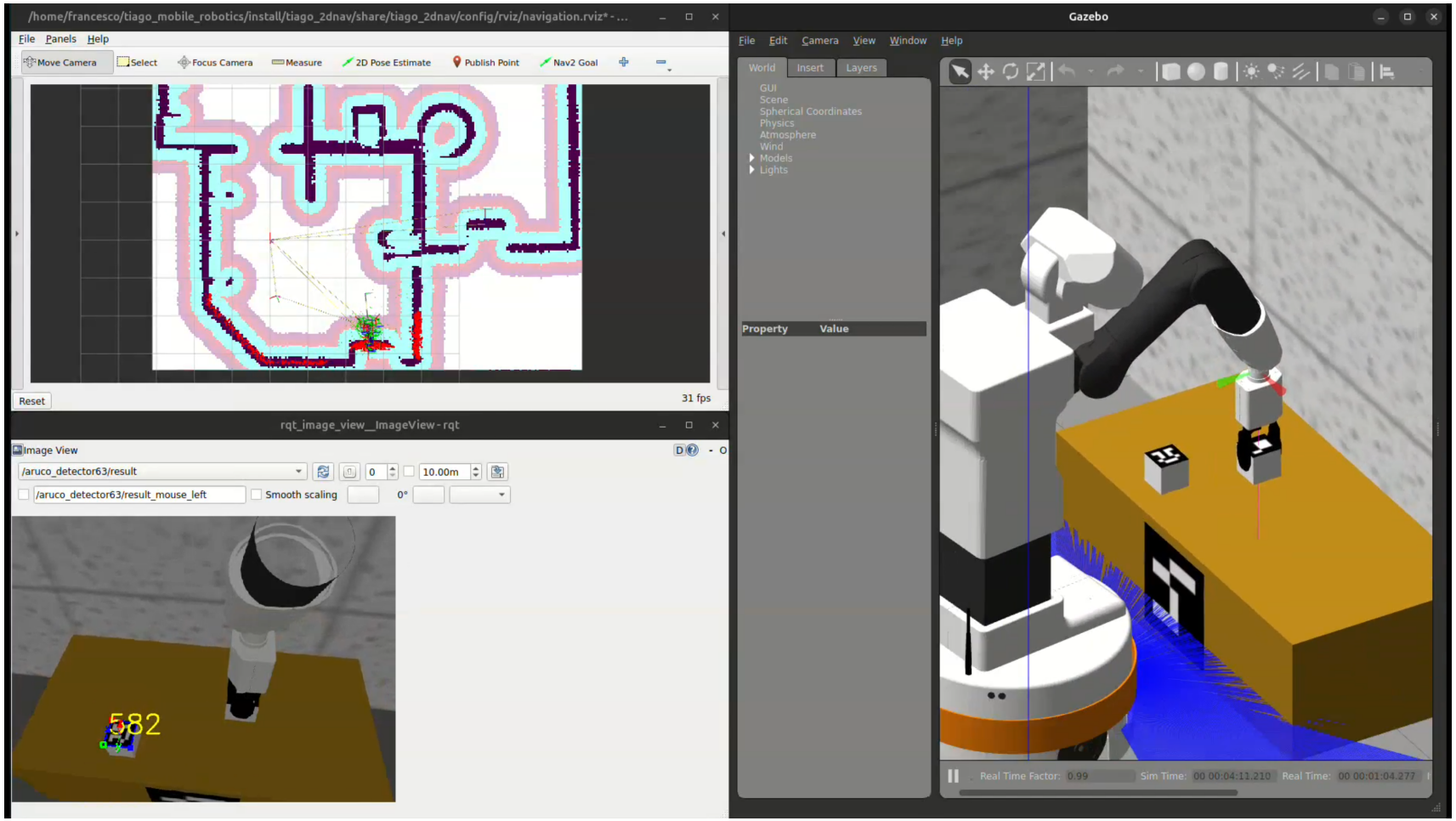


Figure 25: Manipulation Task Execution (Click to play in browser)